STAT 348 Sampling Techniques

Univ. of Saskatchewan, 2026-09

Description

Theory and applications of sampling from finite populations. Includes: simple random sampling, stratified random sampling, cluster sampling, systematic sampling, probability proportionate to size sampling, and the difference, ratio and regression methods of estimation.

Prerequisites

- STAT 242, or STAT 245, or STAT 246

Instructor

- Longhai Li, Professor
- Department of Mathematics and Statistics, University of Saskatchewan
- Email: longhai.li@usask.ca.

Times and Places

- **Lectures:** MWF 12:30 - 1:20, **Arts Building 208**
- **Office Hours:** TBA with Students

Textbook and Course Materials

Textbooks are not required, but you are advised to have one of the following two books:

- **Recommended Text 1:** *Sampling: Design and Analysis*, 2nd Edition, by Sharon L. Lohr (Brooks/Cole). We will roughly cover materials in Chapters 1-6.

- **Recommended Text 2:** *Elementary Survey Sampling*, 7th Edition, by Richard L. Scheaffer, William Mendenhall III, R. Lyman Ott, Kenneth G. Gerow (ISBN-13: 978-0-8400-5361-9). We will roughly cover materials in Chapters 1-9.

- **Lecture Notes:** Available from a shared OneDrive folder with the link given on Canvas (the password is also released on the Canvas page). Assignment questions, solutions, and R code for demonstration are all in this folder.

- **R Code and Spreadsheet Demonstration:** https://longhaisk.github.io/teaching/stat348_rdemo/stat348.html

Computing

We will use RStudio and R for this course.

- **Personal Computer:** Download R, RStudio/Positron/VS-code to your local machine.

- **usask remote computers:** https://teamdynamix.usask.ca/TDClient/33/Portal/KB/Article/116/Access-and-Use-the-USask-Remote-Computer-Lab.

- **Posit Cloud:** (https://posit.cloud/).

- **GitHub Codespaces:** You can also run R and RStudio in the cloud with a GitHub Codespace, without installing anything locally. See https://github.com/codespaces.
- **Google Colab:** You can also run R in the cloud using Google Colaboratory. To open a notebook with R pre-configured, use this direct link: https://colab.research.google.com/#create=true&language=r. Alternatively, you can create a new notebook in Colab and change the runtime type to R (Runtime > Change runtime type > R).

Tentative Schedule / List of Topics

Date	Acad. Week	Topic	Remark
Aug 31	1	1 Introduction to Sampling Techniques, R and R Markdown	**Course Starts (Sep 02)**
Sep 07	2	1 Introduction to Sampling Techniques, R and R Markdown	
Sep 14	3	2 Simple Random Sampling	
Sep 21	4	2 Simple Random Sampling	
Sep 28	5	3 Stratified Sampling	**Assignment 1 due**
Oct 05	6	3 Stratified Sampling	
Oct 12	7	4 Ratio and Regression Estimate	
Oct 19	8	4 Ratio and Regression Estimate	**Midterm**
Oct 26	9	4 Ratio and Regression Estimate	
Nov 02	10	5 Cluster Sampling	**Assignment 2 due**
Nov 09	***N/A***	***—***	***Fall Break – No classes***
Nov 16	11	5 Cluster Sampling	
Nov 23	12	6 Unequal probability sampling	
Nov 30	13	6 Unequal probability sampling	
Dec 07	14	Review	**Assignment 3 due-Course Ends (Dec 07)**

! Important

The schedule is only for reference and may change depending on the actual class pace. The exact assignment due dates and the midterm test date are given on Canvas in the "Assignments" section.

Learning Outcomes

After completing this course, students are expected to grasp the following knowledges and skills:

Topic	Knowledge	Skills	Perc
1 Basic Concepts of Sam-	Understand basic principles of survey design, and identify potential sources	Critically evaluate survey designs, diagnose sources of selection bias and	10%

Topic	Knowledge	Skills	Perc
pling Survey, Selection Bias, Measurement Error	of selection bias and measurement error.	measurement error, and assess their impact on findings.	
2 Simple Random Sampling	Grasp the theoretical foundation, properties, and variance estimation of simple random sampling.	Design simple random sampling schemes and compute estimates and confidence intervals for population parameters.	15%
3 Stratified Sampling	Understand the rationale behind stratification and various sample size allocation methods.	Apply stratified sampling, allocate sample sizes optimally across strata, and analyze resulting datasets.	15%
4 Ratio and Regression Estimate	Understand how auxiliary information improves estimation precision through ratio and regression methods.	Compute ratio and regression estimators, evaluate their variances, and compare their relative efficiencies.	25%
5 Cluster Sampling	Understand the cost-efficiency and design effects associated with cluster sampling.	Design cluster sampling schemes and accurately analyze data collected from clustered populations.	15%
6 Unequal probability sampling	Comprehend inclusion probabilities, sampling with varying probabilities, and Horvitz-Thompson estimators.	Implement unequal probability sampling designs and compute unbiased population estimates from the samples.	20%

> **i Note**
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> The percentages represent the weights in the final examination.

Evaluation

Grading Scheme

3 Assignments: 3 x 10%, 1 Term Test: 20%, 1 Final Exam: 50%.

Assignments and Tests

Assignment questions are released in the one-drive folder. You will submit your solutions via Canvas. **If you miss an assignment without proper excuse, the weight will NOT be shifted to the final.** Undergraduate students will be assigned with different assignments and tests.

Assignments

- I will accept late assignments only for three (3) days beyond the due date. The penalty for your delay is 10 percentage points per day of lateness from the value of the assignment (including weekends). **Extensions are only granted in rare instances (notably as a result of family or medical emergencies) and upon receipt of adequate documentation/proof.**
- Answer the questions in the order they appear in the assignment. Neatness is important.
- Solutions to problems are to be included. Hence, simple answers without work will receive few (or no!) marks.

- Most problems in statistics have a “real-life” basis. Hence, solutions should include not only numerical solutions but also a statement as to what the numbers say about the problem.
- The work handed in must not be an exact duplicate of others.
- Submitting Assignments: The assignment can be typed and/or handwritten. Save your assignment as **one PDF file** (for handwritten assignments, feel free to take a picture/scan of your work and save it as one PDF file). Upload the **PDF file** as an assignment submission in Canvas.
- More details will be provided ahead of each assignment.
- Due Date: See Course Schedule.

Midterm

- The midterm is given in class period.
- Midterms must be written on the dates scheduled. Students must do midterms completely on their own. More details (including syllabus) will be provided ahead of each midterm.
- Type: Short-answer questions, problem-solving, open-book.
- Calculator: A scientific calculator is allowed.
- Make-up exam will not be given. If you miss an exam for a legitimate reason (e.g., illness, emergency) and notify me within 48 hours of the scheduled exam, the weight of the missed exam will be transferred to the final exam.

Final Exam

- Scheduling: Final examinations may be scheduled at any time during the examination period; students should therefore avoid making prior travel, employment, or other commitments for this period. If a student is unable to write an exam through no fault of their own for medical or other valid reasons, documentation must be provided and an opportunity to write the missed exam may be given. Students are encouraged to review all examination policies and procedures: http://students.usask.ca/academics/exams.php.
- The final exam will cover material of the entire course. More details will be provided ahead of the exam.
- Length: 3-hour in-person exam.
- Type: Short-answer questions, problem-solving, open-book.

Criteria That Must Be Met to Pass

The **final exam is a required component of the course**. Students must complete the final exam in order to be eligible to receive a passing grade in this class.

Attendance Expectation

Attendance is highly correlated with student performance. While a syllabus and suggested readings are provided, it is not an adequate substitute for attending class. Your **attendance is highly recommended** but not required, and you will not be graded on your attendance.

Recording of the Course

Recording of the lectures will only be allowed in certain circumstances. Please see the instructor for information on how to receive approval. In general, there will be no videos available for in-person lectures. Therefore, **attendance is strongly recommended**.

Use of Generative AI and Electronic Devices

- AI for Learning vs. Assessment. Students are free (and **encouraged**) to use Generative AI tools as a study aid to understand course concepts, debug code, or explain complex theorems. However, **all submitted work for assignments must be your own.** You must write your own solutions. Directly copying text, derivations, or code from an AI tool and submitting it as your own may receive a **severe penalty** (up to receiving a 0% on the assignment).
- Electronic Devices During Tests. All term tests and the final exam are **Open Book**, meaning you may bring printed notes, textbooks, and lecture slides.
  - **No Electronic Devices:** You are **NOT allowed** to use laptops, tablets, smartwatches, or any other electronic devices during the exam.
  - **Phone Exception:** You are permitted to bring a smartphone, but it must remain stowed away during the writing period. It may **only** be used at the very end of the exam for the specific purpose of taking photos of your answer sheets for submission (if required). Using the phone for any other reason during the exam will be treated as academic misconduct.